

# Positive LDO (LT3042): Working Document

Digital Audio Player • Rev 7

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## 1. Summary

| Field              | Value                                                  |
|--------------------|--------------------------------------------------------|
| Manufacturer / MPN | Analog Devices, LT3042EDD#TRPBF (package to confirm)   |
| Package            | 10-lead DFN 3 × 3 mm, exposed pad (pin 11)             |
| Function           | Ultralow-noise positive LDO, +3V9 → +3.3 V analog rail |

## 2. Design Requirements

| Requirement          | Value                                               |
|----------------------|-----------------------------------------------------|
| Input                | +3V9 from LTC3440, 3.827 to 3.981 V                 |
| Input ripple         | 7.7 mV pp nominal, ±90 mV load step (min 3.74 V)    |
| Output               | 3.32 V                                              |
| Load                 | I/V, summing, OPA1622 (V+ side), DAC analog TBD     |
| Maximum audio output | 0.6 V RMS (EW300: 119 dB/V worst nozzle, 28 Ω ±15%) |
| Output current       | ≈ 27 mA quiescent; ≈ 98 mA peak (200 mA rated)      |
| Current limit        | 125 mA programmed ( $R_{ILIM}$ )                    |
| UVLO                 | 3.5 V rising                                        |
| Power good           | ≈ 91% of $V_{OUT}$ , to MCU GPIO                    |
| Constraints          | Low noise · Small footprint                         |

### 3. Pin Configuration

| Pin  | Name        | Connection                                                 |
|------|-------------|------------------------------------------------------------|
| 1, 2 | IN          | +3V9, $C_{IN}$                                             |
| 3    | EN/UV       | $R_{EN2}$ from IN, $R_{EN1}$ to GND                        |
| 4    | PG          | $R_{PU}$ to +3V3, MCU GPIO                                 |
| 5    | ILIM        | $R_{ILIM}$ to GND (Kelvin to pin 8)                        |
| 6    | PGFB        | $R_{PG2} / R_{PG1}$ midpoint                               |
| 7    | SET         | $R_{SET} \parallel C_{SET}$ to GND; $C_{SET}$ anode to SET |
| 8    | GND         | GND                                                        |
| 9    | OUTS        | OUT at load                                                |
| 10   | OUT         | +3V3A, $C_{OUT}$ , $R_{PG2}$                               |
| 11   | Exposed pad | GND                                                        |

### 4. External Components

| Ref        | Value / Rating                                                                                                   | MPN                       | AL-PN  |
|------------|------------------------------------------------------------------------------------------------------------------|---------------------------|--------|
| $R_{SET}$  | 33.2 k $\Omega$ 0.1% 0402                                                                                        | YAGEO RT0402BRD0733K2L    | #1-085 |
| $C_{SET}$  | 2.2 $\mu$ F $\pm$ 20% tantalum, 16 V MnO <sub>2</sub> , 0603, polarized (no DC-bias derating, not piezoelectric) | Vishay 298D225X0016M2T    | #1-098 |
| $C_{IN}$   | 10 $\mu$ F 10 V X7R 0805 (6.76 $\mu$ F effective at 3.9 V)                                                       | Murata GCM21BR71A106KE22K | #1-081 |
| $C_{OUT}$  | 10 $\mu$ F 10 V X7R 0805 (7.56 $\mu$ F effective at 3.3 V)                                                       | Murata GCM21BR71A106KE22K | #1-081 |
| $R_{EN1}$  | 49.9 k $\Omega$ 0.1% 0402                                                                                        | YAGEO RT0402BRD0749K9L    | #1-138 |
| $R_{EN2}$  | 90.9 k $\Omega$ 0.1% 0402                                                                                        | YAGEO RT0402BRD0790K9L    | #1-084 |
| $R_{PG1}$  | 100 k $\Omega$ 0.1% 0603                                                                                         | YAGEO RT0603BRD07100KL    | #1-066 |
| $R_{PG2}$  | 909 k $\Omega$ 1% 0402                                                                                           | YAGEO RC0402FR-07909KL    | #1-083 |
| $R_{ILIM}$ | 1.0 k $\Omega$ 0.1% 0402                                                                                         | YAGEO RT0402BRD071KL      |        |
| $R_{PU}$   | 10 k $\Omega$ 0603                                                                                               | Generic                   |        |

### 5. Equations and Results

#### Output voltage

$$V_{OUT} = I_{SET} R_{SET} = 100 \mu A \times 33.2 k\Omega = 3.32 V$$

Source: [LT3042 Product Page and Datasheet \(Analog Devices\)](#), SET pin.

#### Undervoltage lockout ( $I_{EN}$ term neglected)

$$V_{IN(UVLO)} = 1.24 \left( 1 + \frac{R_{EN2}}{R_{EN1}} \right) + I_{EN} R_{EN2} = 1.24 \left( 1 + \frac{90.9}{49.9} \right) = 3.499 V$$

Source: [LT3042 Product Page and Datasheet \(Analog Devices\)](#), EN/UV pin.

**Power-good threshold ( $I_{PGFB}$  term neglected)**

$$V_{OUT(PG)} = 0.3 \left( 1 + \frac{R_{PG2}}{R_{PG1}} \right) + I_{PGFB} R_{PG2} = 0.3 \left( 1 + \frac{909}{100} \right) = 3.027 \text{ V (91.2\%)}$$

Source: [LT3042 Product Page and Datasheet \(Analog Devices\)](#), PGFB pin.

**Current limit (ILIM threshold 300 mV; 125 mA·kΩ scale factor)**

$$I_{LIM} = \frac{125 \text{ mA} \cdot \text{k}\Omega}{R_{ILIM}} = \frac{125 \text{ mA} \cdot \text{k}\Omega}{1.0 \text{ k}\Omega} = 125 \text{ mA} \quad (1.27 \times \text{ the 98 mA peak})$$

Source: [LT3042 Product Page and Datasheet \(Analog Devices\)](#), ILIM pin.

**SET noise filter corner (1 μF effective)**

$$f_c = \frac{1}{2\pi R_{SET} C_{SET}} = \frac{1}{2\pi \times 33.2 \text{ k}\Omega \times 1 \mu\text{F}} = 4.8 \text{ Hz}$$

Source: [LT3042 Product Page and Datasheet \(Analog Devices\)](#), SET pin.

**Soft-start (fast start-up to  $\approx 2.7 \text{ V}$ , then the  $R_{SET} C_{SET}$  tail; 1 μF effective)**

$$\tau = R_{SET} C_{SET} = 33.2 \text{ ms}, \quad t_{99} = \tau \ln \frac{0.62}{0.033} = 2.93 \tau \approx 97 \text{ ms}/\mu\text{F}$$

Source: [LT3042 Product Page and Datasheet \(Analog Devices\)](#), fast start-up.

$C_{SET}$  sweep (simulated, LTC3440 replaced by ideal 3.9 V source, 29 Ω load):

| $C_{SET}$ (effective) | $t_{99}$ | Noise, 20 Hz to 20 kHz |
|-----------------------|----------|------------------------|
| 0.47 μF               | 46 ms    | 1.16 μV RMS            |
| 1 μF                  | 98 ms    | 0.63 μV RMS            |
| 2.2 μF                | 214 ms   | 0.40 μV RMS            |
| 4.7 μF                | 455 ms   | 0.34 μV RMS            |

**Headroom**

$$V_{IN(MIN)} - V_{OUT} = 3.74 - 3.32 = 0.42 \text{ V}$$

**Maximum SPL (0.6 V RMS, EW300 copper nozzle 119 dB/V)**

$$SPL_{MAX} = 119 + 20 \log_{10}(0.6) = 114.6 \text{ dB} \quad (116.6 \text{ dB steel nozzle})$$

Load current (0.6 V RMS = 0.85 V peak design maximum, both channels in phase):

| Case                                 | Per channel pk | Signal total | + Quiescent | LT3042 pk               |
|--------------------------------------|----------------|--------------|-------------|-------------------------|
| Full scale, 23.8 Ω (28 Ω -15%)       | 35.7 mA        | 71.3 mA      | 27 mA       | $\approx 98 \text{ mA}$ |
| Full scale, 28 Ω nominal             | 30.3 mA        | 60.6 mA      | 27 mA       | $\approx 88 \text{ mA}$ |
| 110 dB SPL peak (0.35 V RMS), 23.8 Ω | 21.1 mA        | 42.2 mA      | 27 mA       | $\approx 69 \text{ mA}$ |

## 6. Important Values

| Item                   | Value                                                                                   |
|------------------------|-----------------------------------------------------------------------------------------|
| SET pin current        | 100 $\mu$ A                                                                             |
| EN/UV rising threshold | 1.24 V                                                                                  |
| PGFB threshold         | 300 mV                                                                                  |
| Dropout                | 350 mV typ at 200 mA                                                                    |
| Rated output current   | 200 mA                                                                                  |
| Current limit          | 125 mA programmed; ILIM also monitors output current (1:400, 0 to 300 mV)               |
| Peak load              | $\approx$ 98 mA at 0.6 V RMS max (49% of rating)                                        |
| Minimum headroom       | 0.42 V                                                                                  |
| UVLO divider current   | $\approx$ 28 $\mu$ A from +3V9                                                          |
| PG divider current     | $\approx$ 3.3 $\mu$ A from OUT                                                          |
| Fast start-up knee     | $\approx$ 2.7 V (sim, independent of $C_{SET}$ )                                        |
| Output settled (99%)   | $\approx$ 214 ms after enable (2.2 $\mu$ F, sim sweep)                                  |
| Output noise           | 0.40 $\mu$ V RMS, 20 Hz to 20 kHz (2.2 $\mu$ F, sim sweep)                              |
| $C_{SET}$ leakage      | 0.5 $\mu$ A max at 16 V rating $\rightarrow \leq$ 0.5% output error; far lower at 3.3 V |
| Firmware delay         | Wait $\approx$ 214 ms after enable before OPA1622 enable (2.2 $\mu$ F)                  |
| PG in shutdown         | High-Z: reads good while EN/UV low (sim)                                                |

## 7. Open Problems

| # | Problem                                                                                                                              |
|---|--------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Max dropout at peak load not verified; ILIM scale-factor tolerance: confirm minimum limit stays above 98 mA; confirm $R_{ILIM}$ MPN. |
| 2 | EN/UV threshold tolerance, hysteresis and $I_{EN}$ not included.                                                                     |
| 3 | $I_{PGFB}$ error term not included.                                                                                                  |
| 4 | Quiescent (27 mA) unconfirmed: I/V estimate, OPA1622 $I_Q$ at $\pm 3.3$ V unread.                                                    |
| 5 | DAC analog supply allocation to this rail open.                                                                                      |
| 6 | Confirm package (DFN vs MSOP-10E).                                                                                                   |
| 7 | Sim with LTC3440: 27 $\leftrightarrow$ 98 mA load step, 1.2 MHz ripple at OUT (FFT).                                                 |
| 8 | Verify simulated noise against datasheet noise vs $C_{SET}$ curve.                                                                   |

## 8. Revision Log

| Rev | Date       | Change                                                                                                                                                                                                   |
|-----|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0   | 2026-09-21 | Initial issue: $R_{SET}$ , UVLO, PG, soft-start, load current, open problems.                                                                                                                            |
| 1   | 2026-09-22 | $C_{SET}$ 2.2 $\mu$ F nominal ( $\geq 1$ $\mu$ F effective); startup and noise sweeps added.                                                                                                             |
| 2   | 2026-09-22 | $C_{IN} / C_{OUT}$ to 10 $\mu$ F GCM21BR71A106KE22K; derating item closed.                                                                                                                               |
| 3   | 2026-09-22 | Library parts assigned: #1-081, #1-083, #1-084, #1-085. $R_{PG2}$ to 1%.                                                                                                                                 |
| 4   | 2026-09-22 | Max output set to 0.6 V RMS (EW300); peak load 205 $\rightarrow$ 98 mA. Closed: 200 mA rating exceeded.                                                                                                  |
| 5   | 2026-09-22 | Current limit programmed to 125 mA: $R_{ILIM}$ 1.0 k $\Omega$ .                                                                                                                                          |
| 6   | 2026-09-23 | $C_{SET}$ to 2.2 $\mu$ F tantalum 298D225X0016M2T (#1-098), anode to SET. Closed: $C_{SET}$ MPN, firmware delay.                                                                                         |
| 7   | 2026-10-08 | Reformatted to the project document style; linked contents and sources. $R_{EN1}$ 50 k $\Omega$ 0603 (#1-075) $\rightarrow$ 49.9 k $\Omega$ 0402 RT0402BRD0749K9L (#1-138); UVLO recalculated (3.499 V). |

## 9. Sources

1. [LT3042 Product Page and Datasheet \(Analog Devices\)](#)